Project Design Phase-II

Data Flow Diagram & User Stories

Date	07 Feb 2026
Team ID	LTVIP2026TMIDS84828
Project Name	Financial AI Analyzer – Transforming Complex Data into Actionable Insights
Maximum Marks	4 Marks

Data Flow Diagrams

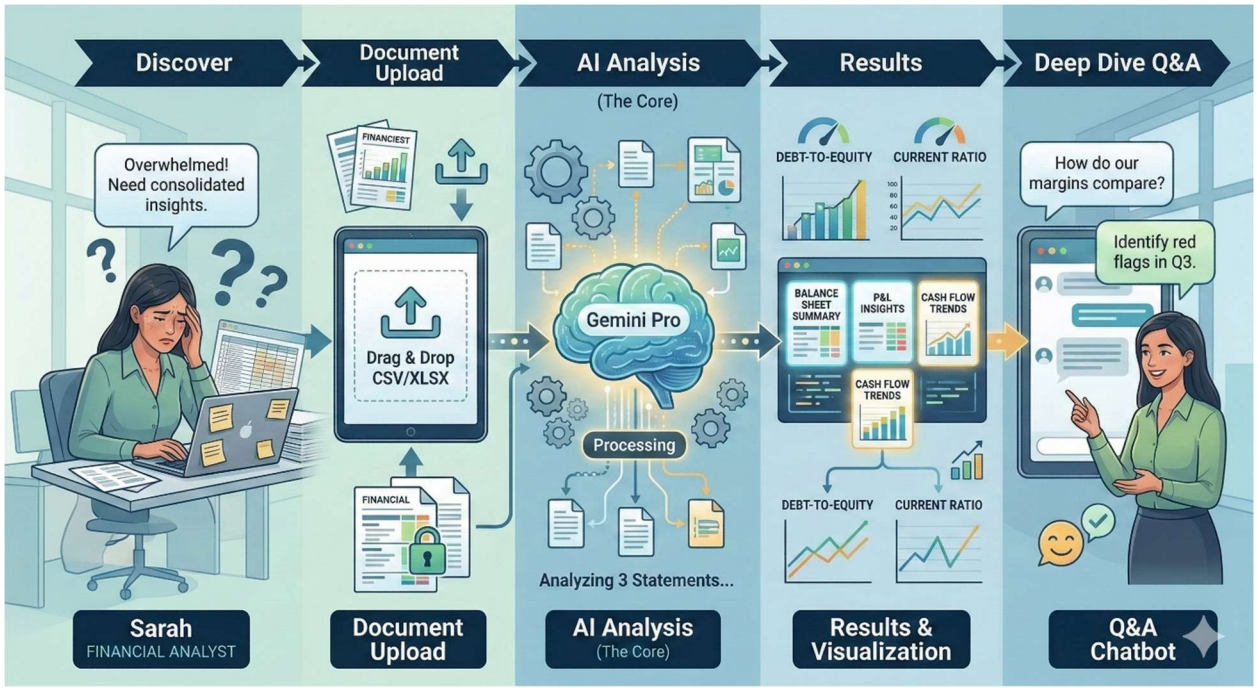
A Data Flow Diagram (DFD) is a graphical representation that illustrates how data moves through a system. It shows the flow of information from input sources to processing units and finally to outputs or storage. DFDs help in understanding how a system processes data and how different components interact with each other.

In the Financial AI Analyzer system, users upload financial datasets or reports, which are then processed by an AI-based analysis engine. The system interprets the financial data, identifies patterns, and generates meaningful insights through visual dashboards and reports. The processed results are then presented to the user in an easy-to-understand format.

The diagram also shows how data flows between the user interface, AI analysis module, financial data processing system, and result dashboard. This structure ensures that financial data is analyzed efficiently and securely while providing accurate insights for better decision-making.

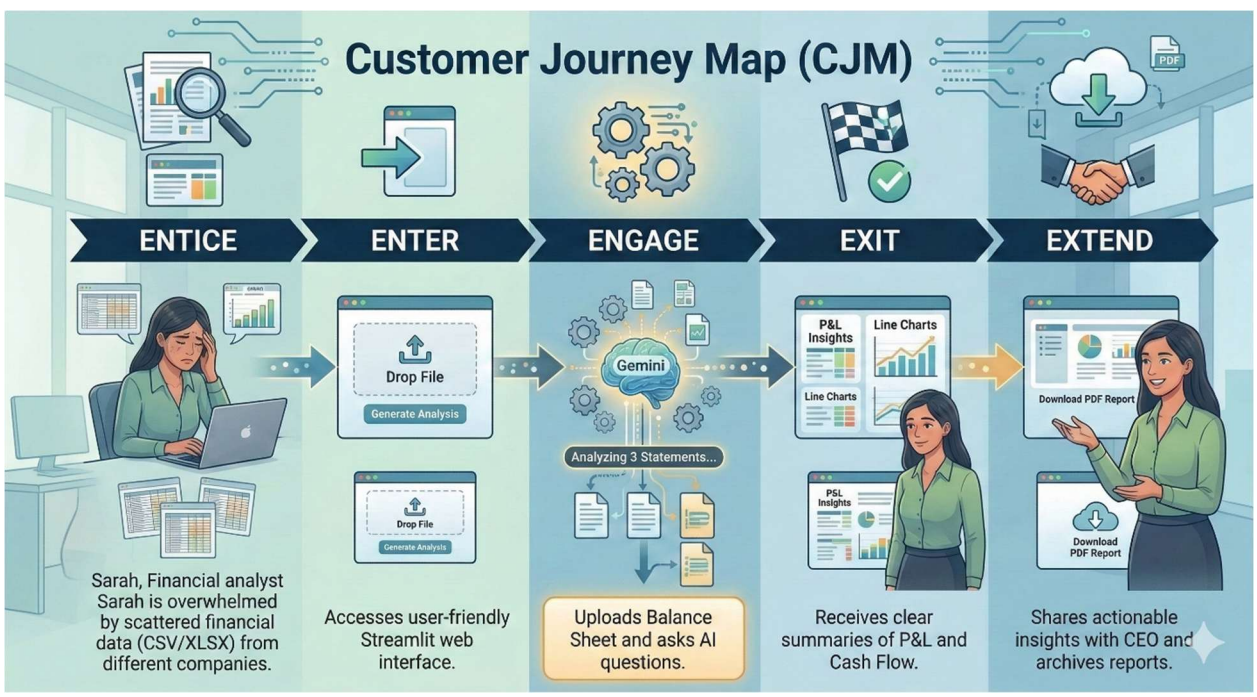
1. Financial Customer Journey Map (CJM)

This follows the "Entice, Enter, Engage, Exit, Extend" model from your design documents, showing a financial analyst's workflow instead of a driver's.



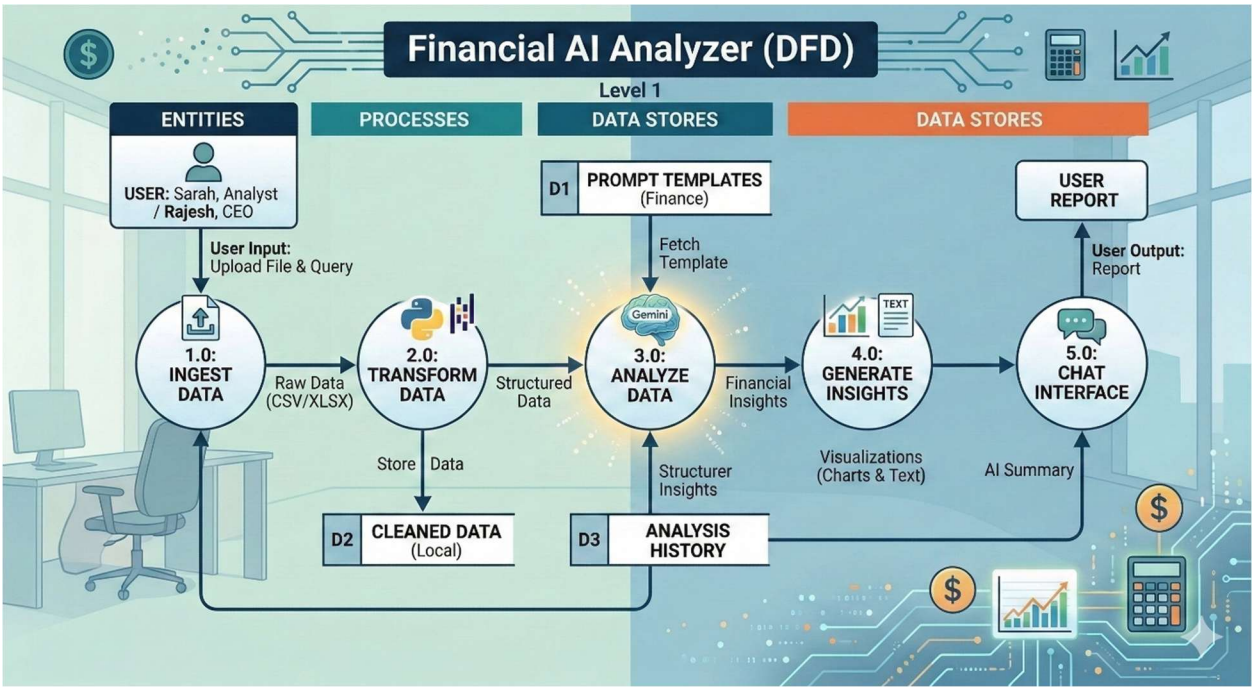
2. Financial Data Flow Diagram (DFD Level 1)

This diagram represents the "traditional visual representation of information flows" as specified in your design documents, showing data moving through your specific technical stack (Streamlit, Pandas, Gemini).



3. Solution Architecture - Financial Decoder

This mirrors the "Client-AI Architecture Model" found in your Technology Stack and Solution Architecture documents, detailing the four layers of your proposed solution.



<u>User Stories:</u>

User Type: Customer (Web User)

Functional Requirement (Epic): Financial Data Analysis

User Story No	User Story / Task	Acceptance Criteria	Priority	Release
USN-1	As a user, I can upload financial datasets in CSV, Excel, or PDF format.	File is successfully uploaded and previewed on the screen.	High	Sprint-1
USN-2	As a user, I can analyze uploaded financial data using AI.	System generates financial insights, charts, and summary reports.	High	Sprint-1
USN-3	As a user, I receive a validation message if no dataset is uploaded.	System shows “No file uploaded” error message.	Medium	Sprint-1

User Type: Customer (Web User)

Functional Requirement (Epic): AI Financial Insights

User Story No	User Story / Task	Acceptance Criteria	Priority	Release
USN-4	As a user, I can ask financial questions using natural language.	AI assistant generates structured financial insights.	High	Sprint-1
USN-5	As a user, I can request investment or financial performance analysis.	System provides summarized financial performance analysis.	High	Sprint-1
USN-6	As a user, I can compare different financial metrics or datasets.	System displays comparison charts and insights.	High	Sprint-1
USN-7	As a user, I can receive financial trend predictions.	AI provides future financial predictions using historical data.	Medium	Sprint-1
USN-8	As a user, I can receive financial risk alerts and recommendations.	System generates risk alerts and suggested financial strategies.	Medium	Sprint-1

User Type: Customer (Web User)

Functional Requirement (Epic): Session Management

User Story No	User Story / Task	Acceptance Criteria	Priority	Release
USN-9	As a user, I can switch between data analysis and AI chat assistant mode.	System switches modes smoothly without losing session data.	**High**	**Sprint-1**
USN-10	As a user, I can view previous analysis results during the session.	Previous analysis reports remain visible until session ends.	**Medium**	**Sprint-1**

User Type: Customer (Web User)

Functional Requirement (Epic): Error Handling

User Story No	User Story / Task	Acceptance Criteria	Priority	Release
USN-11	As a user, I receive a clear error message if the AI fails to generate insights.	System shows a meaningful error message instead of crashing.	**Medium**	**Sprint-1**
USN-12	As a user, I receive a message if the system cannot process the dataset.	System informs the user about unsupported file format or processing error.	**Medium**	**Sprint-1**